

Institutional Coordination under Interdependence: Behavioral Heterogeneity and Scarce Coordination Capacity

Aaron Siebert¹

¹University of Konstanz

August 14, 2026

Abstract

Resource-interdependent agents may control different inputs while requiring complementary bundles to develop. This paper studies how alternative exchange protocols interact with heterogeneous behavior in a stylized five-agent simulation and whether a simple tabular baseline planner can exploit state information when stronger coordination is made scarce. A fixed experiment crossing two build modes, seven behavioral populations and eleven exchange protocols shows substantial package-level differences. Behavioral policy packages alter both development priorities and exchange decisions, while the exchange protocols differ in which additional offer, acceptance, contribution and matching decisions they preserve or bypass. Because unrestricted central clearing performs strongly when it is freely available, I then construct a synthetic coordination-capacity benchmark in which stronger mechanisms consume a regenerating budget. A tabular Q-learning policy trained in this environment does not show a statistically distinguishable welfare advantage over uniform random feasible institutional choice at the reported Monte Carlo precision. Ex-post sequencing diagnostics lead to the same cautious conclusion. Counterfactual rollouts identify continuation-dependent differences among feasible first actions, but they do not establish a generally reliable state-specific deployment rule. The results therefore characterize institutional protocol packages inside the model and caution against inferring useful adaptive timing from structured policy behavior when the learned state-to-action mapping does not improve welfare over simpler multi-institution choice.

Introduction

Economic activity often depends on interactions between actors who do not control all resources or capabilities required to achieve their objectives. When exchange is not frictionless, the rules governing these interactions become part of the allocation problem itself. Institutional economics emphasizes that institutions structure economic interaction and incentives, while transaction costs can affect which arrangements for exchange and coordination are workable [North, 1990, Coase, 1960].

The simulation studied here represents this problem in an intentionally simple form. Five agents have asymmetric access to resources but require complementary bundles to develop, so a missing input can block an otherwise feasible project. Outcomes may also depend on behavioral composition: experimental and behavioral economics show that individuals can differ in fairness concerns, reciprocity, relative-outcome preferences and willingness to cooperate [Fehr and Schmidt, 1999, Bolton and Ockenfels, 2000, Charness and Rabin, 2002, Fischbacher et al., 2001].

Exchange protocols provide a second source of variation. They determine how transfers are matched, which decisions remain voluntary and how strongly allocation

is coordinated. Mechanism-design research motivates treating allocation rules as objects of analysis rather than a fixed background condition [Hurwicz, 1973, Maskin, 2008]. The protocols studied here are much simpler and each combines several operational rules, so the comparisons concern complete protocol packages rather than an isolated causal effect of centralization.

Agent-based simulation provides a controlled way to study these interactions. Heterogeneous agents and institutional rules can be specified explicitly and their system-level consequences observed under repeated simulation [Tesfatsion, 2006, Epstein, 1999]. The present model is therefore used as a testbed rather than as an estimate of a particular real-world institution.

Recent reinforcement-learning work shows that social planners and allocation mechanisms can be trained in simulated behavioral environments [McKee et al., 2023, Koster et al., 2025]. A related review of deep mechanism design emphasizes that such approaches depend on how human preferences are modeled and on how mechanisms are aligned with potentially diverse groups [Tacchetti et al., 2025]. The present study takes a narrower computational approach by varying stylized behavioral policy populations explicitly and by testing whether a

simple planner can select among pre-specified exchange protocols when stronger coordination is scarce.

The study proceeds in two stages. First, a three-factor experiment crosses two build modes, seven behavioral policy populations and eleven fixed exchange protocols, supplemented by two behavioral-composition sweeps. Second, because unrestricted strong coordination leaves little sequential choice, I introduce a synthetic coordination-capacity constraint under which stronger mechanisms consume a regenerating budget. This creates an intertemporal decision problem in which using costly coordination now can restrict later options.

A simple tabular Q-learning planner is then trained to choose among seven exchange mechanisms using a compact description of the simulated economy. Its objective combines average score with the score of the weakest-performing agent and its performance is compared with multi-institution baselines and rollout diagnostics on evaluation seeds that are disjoint from training.

The central research questions are:

*When resource-interdependent agents follow heterogeneous decision policies, how do alternative exchange protocols affect development?
Can a simple tabular baseline planner exploit state information when stronger coordination is made scarce?*

The results show large differences among protocol-behavior packages and an in-model output-distribution trade-off between standard and equity-oriented central matching. In the adaptive extension, the trained Q-policy does not show a statistically distinguishable welfare advantage over uniform random feasible choice at the reported precision, while rollout diagnostics show continuation-dependent first-action differences without identifying a general deployment rule.

Literature

The model combines ideas from institutional economics, behavioral economics, agent-based modeling and reinforcement learning. These literatures motivate the questions and modeling choices. They do not calibrate the simulation’s numerical parameters.

Institutions, Coordination and Resource Allocation North treats institutions as constraints that structure interaction and incentives, while Coase emphasizes that exchange and coordination are not costless and that transaction costs affect workable organizational arrangements [North, 1990, Coase, 1960]. Together, these perspectives motivate treating the exchange protocol as part of the economic environment rather than assuming frictionless transfers.

Mechanism design makes the allocation rule itself an object of analysis by asking how rules map individual actions or information into outcomes [Hurwicz, 1973, Maskin, 2008]. The protocols used here do not satisfy the informational or equilibrium requirements of formal implementation theory, but this perspective motivates comparing alternative allocation procedures explicitly.

Research on collective action also cautions against assuming a universal ranking between centralized and decentralized organization. Ostrom shows that self-governed institutions can organize common-pool resources under conditions where simple market or external-control accounts are incomplete, while experimental work with Walker and Gardner shows that communication and internally available sanctioning arrangements can change cooperative outcomes [Ostrom, 1990, Ostrom et al., 1992]. These settings differ from the exchange economy studied here, but they motivate treating institutional performance as conditional on behavior and rules.

Experimental institutional research sharpens that point: Kosfeld, Okada and Riedl show that groups can endogenously form sanctioning institutions, linking institutional outcomes to participants’ willingness to support them [Kosfeld et al., 2009]. Fehr, Powell and Wilkening demonstrate that theoretically attractive implementation mechanisms can face behavioral constraints in practice [Fehr et al., 2021]. The present study does not reproduce these institutions. It carries forward the narrower idea that outcomes depend on the interaction between rules and behavioral responses.

Behavioral Heterogeneity Behavioral economics provides several reasons not to impose one representative decision rule on all agents. Fehr and Schmidt model inequity aversion, while Bolton and Ockenfels formalize concern for relative standing and equity [Fehr and Schmidt, 1999, Bolton and Ockenfels, 2000]. Charness and Rabin provide evidence that social welfare, outcomes of worse-off participants and reciprocity can influence decisions [Charness and Rabin, 2002]. Fischbacher, Gächter and Fehr further show heterogeneity within collective-action settings, including conditional cooperation and freeriding behavior [Fischbacher et al., 2001].

These findings motivate heterogeneous policy profiles, not the specific numerical profiles used here. Need-based, cooperative, selfish, hoarding, competitive and fairness-sensitive agents are stylized rule systems within one common utility-based architecture. Their weights affect project targeting and exchange valuation, while separate parameters govern offers, acceptance, reserves, payment limits and pool contributions. The mixed population combines five policy classes. The experiment therefore asks how complete behavioral policy packages interact with exchange protocols while holding the underlying resource environment fixed.

Agent-Based Computational Modeling Agent-based models are useful when aggregate outcomes emerge from repeated interaction among heterogeneous actors. Tesfatsion develops agent-based computational economics as a constructive approach to dynamic economic systems [Tefatsion, 2006], while Epstein’s generative perspective similarly asks whether specified micro-level rules can produce system-level patterns of interest [Epstein, 1999]. This approach does not establish empirical validity by itself, but it allows behavioral populations and exchange protocols to be changed systematically while the remaining model mechanics are held fixed. That controlled comparison is the main reason for using simulation here.

Reinforcement Learning and Adaptive Institutional Choice The adaptive extension turns institutional choice into a sequential problem: using coordination capacity in one round can restrict feasible actions later. Q-learning estimates action values from repeated state–action–reward transitions without requiring an explicit transition model, while reinforcement learning more generally formalizes policy learning from repeated interaction with an environment [Watkins and Dayan, 1992, Sutton and Barto, 2018]. This makes Q-learning a useful simple baseline for testing whether the planner’s observed state contains exploitable information about when stronger coordination should be used.

Two limitations of that state representation are directly relevant to interpretation. Jaakkola, Singh and Jordan study reinforcement learning when the learner has only partial information about the underlying state [Jaakkola et al., 1994]. Here the planner does not observe the full agent state or behavioral scenario. Singh, Jaakkola and Jordan study reinforcement learning with compact state aggregation and analyze how aggregation affects learning [Singh et al., 1994]. Here many underlying situations are compressed into the same discretized state. These papers therefore motivate caution in interpreting the performance of the tabular planner rather than serving as claims that the current representation is optimal.

Recent work brings reinforcement learning directly into institutional design. McKee et al. [2023] train a deep-RL social planner in simulation to restructure interaction networks in a cooperation game and then evaluate it with human groups while Koster et al. [2025] use deep RL to learn state-dependent resource-allocation mechanisms in an iterated common-pool resource setting, training against models of human behavior and subsequently evaluating the resulting mechanism with human participants. Tacchetti et al. [2025] review recent applications of reinforcement learning to mechanism design, including learned auction, taxation and redistribution rules. They emphasize that such approaches depend on how human behavior is represented in the training environment and on how potentially conflicting

preferences are incorporated into the planner’s objective. Both issues are relevant here: behavioral populations are varied explicitly rather than represented by a single agent model and the adaptive planner’s welfare objective is specified as part of the experimental design. The present study nevertheless addresses a narrower computational problem. Rather than learning a new allocation rule, it varies stylized behavioral policy populations explicitly and asks whether a simple planner can select among pre-specified exchange protocols under synthetic coordination scarcity.

Taken together, the literature motivates a controlled computational comparison of exchange protocols under heterogeneous behavioral policies, followed by a deliberately simple adaptive benchmark under scarce coordination capacity.

Methods

Simulation Environment

The model contains five agents, five resource types and 20 rounds. Its resource-exchange structure was loosely inspired by *The Settlers of Catan*, but it does not reproduce the board game or analyze *Catan* play. Agents have asymmetric access to resources while development requires complementary bundles, so an agent can be blocked by one missing input. Trade can remove these local bottlenecks. At the start of each game, agents have empty resource stocks, zero developmental score, zero infrastructure, zero production and advanced-production sites, zero innovation and no active leader bonus. Each agent begins with two resource draws per round. Subsequent production capacity can increase through building.

Each round consists of four stages: resource production, institution-mediated exchange, building and score, capacity and access updates.

Building options include infrastructure, production sites, advanced production and innovation. Each agent may complete at most four builds per round. Build decisions are sequential: after every successful build, the current public state is refreshed before the agent can attempt another build and the sequence stops early if the policy selects no further feasible build. Because successful builds can change future productive capacity, exchange has path-dependent consequences. In the fixed-institution experiments, aggregate development is measured by final total score, while distributional spread is the final max–min score range, the highest minus the lowest individual final score.

Build mode is an additional game-level factor layered on top of the behavioral policies’ common utility-based build system. In both modes, policy-specific utility weights enter project-target and feasible-build scoring, including weights on points, productive ca-

capacity, site progress, innovation and target progress. The development-oriented mode omits the leader bonus-specific incentive term from project and build scoring. The crown-aware mode adds that term, which can redirect choices toward claiming, defending or stealing infrastructure- and innovation-leader bonuses. The underlying leader bonuses themselves remain part of the game environment in both modes.

The Supplementary Materials report the principal resource, build, leader-bonus, institution and adaptive-planner settings.

Behavioral Heterogeneity

The main fixed experiment contains seven behavioral policy populations: need-based, cooperative, selfish, hoarding, competitive, fairness-sensitive and a mixed population. The first six are homogeneous five-agent populations using one policy class throughout. The mixed population contains one agent from each non-need-based policy class: cooperative, selfish, hoarding, competitive and fairness-sensitive. The need-based policy is the separate homogeneous baseline and is not included in the mixed population.

All behavioral policies use the same utility-based decision architecture and the same feasible project set, but they differ in both utility weights and direct exchange parameters. The utility weights affect which development target is valued and which feasible build is chosen. The same target valuations also shape resource requests and trade utility. Additional policy parameters govern preferred reserves, payment limits, offer and acceptance thresholds as well as public pool contributions. Cooperative policies accept weaker exchange gains and place more weight on social welfare, hoarding policies protect larger reserves and place more weight on security, competitive policies emphasize relative advantage and fairness-sensitive policies emphasize equity. Behavioral research motivates studying such heterogeneity [Fehr and Schmidt, 1999, Bolton and Ockenfels, 2000, Charness and Rabin, 2002, Fischbacher et al., 2001], but these weights and thresholds remain stylized design choices. The experiment therefore compares these behavioral policy packages, not estimated behavioral types, while holding the resource environment and feasible project set fixed.

Exchange Institutions

The model compares institutions that differ in how strongly they organize exchange. This comparison follows the general idea that allocation rules can be treated as objects of design and comparison [Hurwicz, 1973, Maskin, 2008], while remaining deliberately simpler than a formal mechanism-design model.

The final adaptive planner chooses among seven options for each individual round played in a game (see Table 1). The action set spans repeated bilateral bar-

Table 1: Institutional mechanisms used by the adaptive planner. The institutions differ in how directly they organize resource transfers and in the modeled coordination capacity required to use them. The table summarizes their conceptual role. Exact implementation details are given in the Supplementary Materials.

Institution	Basic mechanism	Coordination
Bilateral 3-pass	Repeated direct exchange	Low
Clearinghouse	Aggregates compatible exchanges	Moderate
Public pool	Collective round-local pool	Moderate
Subsidized catch-up	Supports bottlenecked agents	Moderate
Central cap	Greedy central matching with cap	High
Equity cap	Capped matching favoring weaker agents	High
Full central	Greedy centralized matching	Highest

gaining, clearinghouse coordination, public pooling, targeted subsidy and progressively stronger central matching. “Full central clearing” is not a complete price-clearing or globally optimal market solution: it is a centralized greedy matcher that constructs one-for-one barter candidates and ranks them by target-shortage reduction. These treatments intentionally change several dimensions at once, including voluntary participation, matching scope, redistribution and throughput. Consequently, a difference between two conditions is a difference between bundled exchange protocols. The design can reveal how those packages behave in the simulated environment, but it does not isolate the causal contribution of any one institutional dimension.

Experimental Design and Simulation Data

All reported data are generated by the simulation. The project therefore does not use an external observational dataset. Instead, experimental comparisons are created by repeatedly running the same model under controlled changes to build mode, behavioral composition, institutional rules and random seeds. This use of simulation follows the agent-based approach described by Tesfatsion [2006] and Epstein [1999]: system-level outcomes are generated from specified agent and interaction rules, after which counterfactual institutional conditions can be compared.

The fixed experiment is a three-factor design crossed with matched simulation seeds:

$$2 \times 7 \times 11 \times 200 = 30,800 \text{ games}$$

The three treatment factors are build mode, behavioral population mix and institution. The 200 seeds serve as matched simulation blocks and replication indices. The seven populations are need-based, cooperative, selfish, hoarding, competitive, fairness-sensitive and mixed. The eleven fixed institutions are no trade, bilateral bargaining, bilateral three-pass, catch-up bilateral, bottleneck-priority bilateral, clearinghouse bargaining, subsidized catch-up, public pool, central clearing, equity-weighted central clearing and central clearing capped at two successful trades per round. Within

each seed, the balanced asymmetric access profiles are reused across conditions, creating matched comparisons.

The two build modes are crossed factorially with these populations and institutions rather than treated as behavioral types. Both modes are included in the full fixed experiment. Where a later comparison requires one build mode, the development-oriented condition is used unless stated otherwise. The comparison between both build modes is reported in the Results section.

A separate composition experiment contains two five-agent gradients: cooperative $\rightarrow$ hoarding and need-based $\rightarrow$ competitive, both under development-oriented building. For restrictive count $k \in \{0, \dots, 5\}$, exactly k of the five agents use the restrictive endpoint policy and the remaining $5 - k$ use the baseline policy. Each composition-institution condition is evaluated over 100 matched seeds.

The standard-versus-equity central comparison reported below uses the development-oriented fixed-experiment rows only. Standard and equity-weighted central clearing are compared within the same seed and behavioral population, giving 1,400 paired differences across 200 seeds. Because each seed appears in all seven populations, the confidence interval is calculated by re-sampling whole seeds rather than treating the 1,400 rows as independent. Each of 10,000 bootstrap draws samples 200 seeds with replacement and carries all seven population differences for a sampled seed together. The 2.5th and 97.5th percentiles of the resulting mean differences form the reported 95% interval.

The same paired logic is used elsewhere when conditions share simulation seeds. The purpose is simple: if a particular seed happens to create an unusually favorable or unfavorable game, both conditions experience that same underlying random environment before their difference is calculated. For the build-mode comparison, crown-aware and development-oriented scores are subtracted within the same seed, behavioral population and institution and the 200 seeds are treated as the independent blocks for uncertainty. For each composition-sweep endpoint comparison, the difference between the $k = 5$ and $k = 0$ populations is first calculated for each of the same 100 seeds. A 10,000-draw bootstrap then re-samples those 100 paired differences with replacement. The 2.5th and 97.5th percentiles form the reported 95% interval. Development-oriented population-institution cell means are reported separately with Student- t 95% confidence intervals across the 200 seed-level games in each cell.

All confidence intervals in this paper quantify stochastic variation induced by repeated simulation under the stated model and fixed parameterization. They are Monte Carlo uncertainty summaries for this synthetic environment. The adaptive experiments introduce scarce coordination capacity and an institutional planner.

Scarce Coordination Capacity

The unconstrained benchmark reveals that the centralized greedy matcher is particularly effective in many modeled populations. If such a mechanism can be selected freely every round, a sequential institutional-choice problem is difficult to sustain because the planner has little reason to hold off on stronger coordination.

The adaptive environment therefore treats coordination capacity as a limited and regenerating resource. This is a synthetic modeling device intended to create an explicit intertemporal trade-off. It should not be interpreted as an empirical measure of administrative, political or state capacity. The particular accounting rule is also substantive: bilateral three-pass acts as a zero-capacity default, stronger mechanisms consume capacity and an action must be able to cover its maximum possible commitment before selection, even though only realized workload is charged afterward. These choices define the benchmark rather than estimate a real governance capacity.

Let G_t denote available coordination capacity in round t . Capacity is initialized at and capped by

$$G_{\max} = 8 \quad (1)$$

and regenerates by

$$g = 1 \quad (2)$$

unit per round.

The realized cost of choosing institution a is

$$c_t(a) = b(a) + s_t(a) + w_t(a) \quad (3)$$

where $b(a)$ is the institution's base coordination burden, $s_t(a)$ is a switching or reconfiguration cost and $w_t(a)$ is a workload cost based on the amount of exchange actually handled.

Because realized workload is not known before the institution is selected, the planner must be able to afford its maximum possible commitment,

$$k_t(a) = b(a) + s_t(a) + w_{\max}(a) \quad (4)$$

An action is selectable only if

$$k_t(a) \leq G_t \quad (5)$$

After the institution operates, only its realized cost is deducted:

$$G_{t+1} = \min(G_{\max}, G_t - c_t(a_t) + g) \quad (6)$$

Bilateral bargaining has no base coordination burden. Intermediate mechanisms carry a base cost of one unit, capped central mechanisms a base cost of two units and full central clearing a base cost of three units. Entering a different non-bilateral institution adds one unit of switching cost. Remaining in the same coordinated

institution does not incur this additional charge and returning to bilateral exchange is free. The exact capacity-cost table is reported in the Supplementary Materials.

These numerical values are design assumptions chosen to make coordination meaningfully scarce over the 20-round horizon. The mixed action patterns reported later should therefore not be interpreted as independent validation of these costs: the environment is intentionally constructed so that stronger coordination cannot simply be used without consequence.

Planner Objective

Planner welfare combines average performance with lower-tail welfare:

$$\Phi_t = (1 - \lambda)\bar{s}_t + \lambda s_t^{\min} \quad (7)$$

where $\bar{s}_t$ is mean score, $s_t^{\min}$ is the score of the weakest-performing agent and

$$\lambda = 0.25 \quad (8)$$

The planner therefore places 75% weight on average performance and 25% on the weakest agent. This objective is a modeling choice. It is intended to prevent the adaptive planner from treating gains to already successful agents as the only relevant outcome.

The reinforcement-learning reward is the change in planner welfare,

$$r_t = \Phi_{t+1} - \Phi_t \quad (9)$$

so actions are rewarded when they increase the selected welfare measure from one round to the next. In the Q-learning update below, γ is the discount factor: it determines how strongly estimated future rewards are weighted relative to the immediate reward. Here $\gamma = 1$, so future rewards are not discounted. Summing the round-by-round rewards over a complete game gives

$$\sum_t r_t = \Phi_T - \Phi_0$$

The intermediate welfare values cancel because each round's Φ_{t+1} is subtracted again in the following reward. Since all agents begin with zero score, $\Phi_0 = 0$, so maximizing total reward over the game is equivalent to maximizing final planner welfare Φ_T .

Q-Learning Planner

The adaptive planner uses tabular Q-learning [Watkins and Dayan, 1992, Sutton and Barto, 2018]. Its state representation compresses the game into seven features describing: remaining coordination capacity, game phase, score-gap category, current bottlenecks, idle productive infrastructure, viability of voluntary exchange and the institution used in the previous round.

This representation intentionally summarizes rather than reproduces the complete simulation state. The

planner does not directly observe every agent's full inventory, target, behavioral history or all possible exchange opportunities. Behavioral composition is hidden except insofar as its consequences appear in the seven features. Time is compressed into four phase bins.

Let $\mathcal{A}(s_t)$ denote the institution set that is feasible under the current coordination capacity. During training, actions are selected using an ϵ -greedy policy:

$$a_t = \begin{cases} \text{random action from } \mathcal{A}(s_t), & \text{with probability } \epsilon, \\ \arg \max_{a \in \mathcal{A}(s_t)} Q(s_t, a), & \text{with probability } 1 - \epsilon \end{cases} \quad (10)$$

Q-values are updated according to

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \gamma \max_{a \in \mathcal{A}(s_{t+1})} Q(s_{t+1}, a) - Q(s_t, a_t) \right] \quad (11)$$

The learner was trained for 12,000 episodes. Each episode samples uniformly from six behavioral scenarios: homogeneous need-based, cooperative or hoarding populations, one mixed population and two transition scenarios in which all five agents switch at the start of simulation round index 10 from need-based to competitive behavior or from cooperative to hoarding behavior. These six scenarios are a selected adaptive training set rather than the full seven-population factorial design: homogeneous selfish, fairness-sensitive and full-game competitive populations are not separate adaptive scenarios. Learning rates decay by state-action visit count from 0.25 to a 0.01 floor and exploration decays quadratically from 1.0 to 0.03 over the first 90% of training. Evaluation disables exploration and learning. Q initialization, tie-breaking, terminal handling and exact schedules are reported in the Supplementary Materials.

The evaluation seeds are disjoint from Q training. The same evaluation set was also used when comparing additional Q-learning variants, so it is held out from training but should not be interpreted as an untouched final test set. Only 21 of 12,000 evaluation decisions (0.175%) encountered a state unseen during Q training. These states had zero-valued read-only Q entries and therefore selected bilateral three-pass through the stable action-order tie-break. This state-level count does not establish adequate state-action coverage: a familiar state can still contain feasible actions that received little training experience.

Policy Evaluation and Baselines

Policy evaluation uses four comparisons with different purposes. Permanent bilateral is a weak fixed reference. Random feasible is the main independent multi-institution baseline and chooses uniformly among currently feasible institutions. Frequency-informed random

feasible and the shuffled learned sequence are diagnostics derived from Q behavior on the evaluation games. Both remove part of the learned state-dependent sequencing rather than providing independent predictive baselines.

All policies are evaluated on the same 100 seeds in each of six scenarios. Paired comparisons therefore hold the simulated environment constant as far as possible, with production randomness separated from policy randomness. For pooled comparisons, all six scenario differences belonging to a seed remain together. A 10,000-draw bootstrap resamples the 100 seed blocks with replacement and the 2.5th and 97.5th percentiles of the resampled mean differences form the reported 95% interval.

Random feasible At each round, an institution is sampled uniformly from the currently feasible set. This provides the simplest multi-institution benchmark and contains no learned state dependence.

Frequency-informed random feasible This is an ex-post diagnostic rather than an independent predictive baseline. Its sampling weights are constructed from the Q-policy’s aggregate action counts on the same fixed evaluation games. At each decision, those counts are renormalized over the currently feasible set. Because earlier actions change remaining coordination capacity, the realized action shares need not exactly reproduce those of the Q-policy. The diagnostic asks what happens when the Q-policy’s evaluation-set marginal action frequencies are used as broad weights while its state-to-action mapping is removed.

Shuffled learned sequence This second ex-post diagnostic starts from the learned Q action sequence generated on the same evaluation seed-scenario game and shuffles that 20-action sequence once within the game. During the new trajectory, queued actions that are infeasible are skipped until a feasible queued action is found. If no usable queued action remains, selection falls back to frequency-informed random feasible choice. Switching costs, workload costs and subsequent feasibility are recalculated normally along the new trajectory. The diagnostic preserves the learned evaluation sequence as much as possible while disrupting its original order.

The main outcome is final planner welfare. Permanent bilateral is retained for scale, while conclusions about learned state dependence rely on random feasible and the two sequencing diagnostics described above.

Counterfactual Branch Diagnostic

A separate rollout diagnostic asks a narrower question: when the simulated environment is frozen at a

checkpoint, do alternative feasible first institutions produce distinguishable downstream welfare under controlled continuation policies? This is a diagnostic of continuation-conditioned first-action value. It does not by itself isolate a pure “use coordination now versus save it for later” timing effect, because the value assigned to the first action depends on the continuation rule that follows it.

The diagnostic uses 18 independent base games, three per behavioral scenario and checkpoints at rounds 3, 10 and 16. Every feasible first institution is evaluated with 40 continuations from a deep-copied state. The first 20 rollouts are used only to identify provisional best and second-best actions. The remaining 20 rollouts then evaluate those already selected actions. This separation matters because using the same noisy rollouts both to choose the apparent winner and to measure its advantage would tend to overstate that advantage: the action with the largest favorable simulation noise would be selected by construction. Candidate branches share exogenous production randomness at the branch point where technically meaningful and then diverge normally.

The diagnostic is repeated under three continuation rules: return to the Q-policy, permanent bilateral continuation and random feasible continuation. Aggregate uncertainty is clustered at the 18-base-game level because the three checkpoints drawn from a base game are not independent. Aggregate 95% intervals use a 10,000-draw percentile cluster bootstrap: base games are sampled with replacement, all checkpoint contrasts belonging to each sampled game are carried together and the 2.5th and 97.5th percentiles form the interval. The checkpoint-level 95% intervals are unadjusted descriptive intervals, no multiple-comparison correction is applied across the checkpoints. These intervals describe the precision available at individual states and are not treated as a formal rank-stability test. The diagnostic remains exploratory because only 18 independent base games underlie the aggregate estimates and the estimated first-action contrasts depend on the continuation rule used after the branch.

Code and reproducibility The simulation code, analysis scripts, stored outputs used in this paper and reproducibility instructions are available in the accompanying GitHub repository: <https://github.com/ASbrt/competition-under-interdependence>.

Results

Bundled Exchange Protocols Produce Large, Behavior-Dependent Differences Across the experiment design, central clearing has the highest mean final total score for every population under both build modes. The pooled build-mode contrast is -0.852 final-score points on average, with a seed-clustered 95% confidence

interval of $[-1.045, -0.655]$. This is small relative to institutional and composition contrasts that reach tens of score points in several conditions. Crown-aware outcomes are lower in 54 of 77 population–institution cells, equal in 12 and higher in 11, but the identity of the top-ranked institution does not change.

The separate composition sweeps show that stable top rankings in the fixed-experiment grid do not imply that institutional performance is insensitive to behavior. Replacing a five-agent cooperative population with hoarding agents reduces bilateral three-pass score by 28.92 points on average (95% CI $[-34.70, -23.21]$) and public-pool score by 110.58 points (95% CI $[-114.16, -107.09]$). In the same cooperative-to-hoarding transition, central clearing instead increases by 17.12 points (95% CI $[12.22, 22.10]$).

The need-based-to-competitive transition produces a different pattern. Bilateral three-pass falls by 41.39 points (95% CI $[-46.99, -35.77]$) and the public pool by 116.51 points (95% CI $[-119.27, -113.76]$), but central clearing also falls by 18.00 points (95% CI $[-22.84, -13.14]$). Capped central coordination declines more moderately by 5.10 points (95% CI $[-9.06, -1.00]$). Thus the cleanest result is not that central institutions are behaviorally invariant. Rather, bilateral exchange and voluntary pooling are strongly composition-sensitive in both sweeps, while centralized mechanisms respond differently to the two behavioral transitions.

The behavioral policies differ in both development and exchange decisions, while the protocols preserve or bypass different exchange decision points. The composition sweeps therefore measure interactions between complete behavioral-policy packages and protocol interfaces rather than an independently identified effect of centralization or exchange willingness. This interpretation is consistent with experimental work showing that institutional outcomes depend on behavioral responses to the rules participants face [Kosfeld et al., 2009, Fehr et al., 2021].

Equity-Oriented Coordination Trades Aggregate Score for a Smaller Final Score Gap The comparison between standard central and equity-oriented central coordination shows a clear difference between aggregate and distributional outcomes (Fig. 2). This comparison uses the development-oriented build mode only, pooling all seven behavioral populations in the fixed experiment and pairing the two institutions within each of 200 seeds. Standard central clearing produces 3.286 more final total-score points on average than equity-oriented central coordination, with a seed-clustered 95% confidence interval of $[1.861, 4.704]$.

At the same time, standard central clearing produces a final max–min score range that is 2.356 points larger on average, with a 95% confidence interval of $[1.868, 2.844]$. The equity-oriented mechanism therefore

gives up some aggregate development while compressing the distance between the strongest- and weakest-scoring agents in this modeled environment.

The max–min score range is deliberately a simple spread measure rather than a complete inequality index. Likewise, the equity mechanism is an explicit modeling choice, not an empirically estimated social-welfare institution. The result should therefore be read as an in-model output–distribution trade-off rather than a general claim about real redistributive institutions. Behavioral theories in which relative and lower-tail outcomes matter provide conceptual motivation for keeping these objectives distinct [Fehr and Schmidt, 1999, Bolton and Ockenfels, 2000, Charness and Rabin, 2002].

Scarce Coordination Produces Selective Institutional Use, but the Action Mix Is Descriptive

Once coordination capacity is limited, stronger mechanisms cannot simply be used continuously. The Q-policy therefore allocates most rounds to cheaper institutions and uses costly coordination selectively (Fig. 3). Across the evaluation set, bilateral three-pass accounts for 57.93% of actions, subsidized catch-up for 18.88%, clearinghouse exchange for 12.87% and full central clearing for 2.58%. Public pooling and the two capped central mechanisms account for the remaining 7.74%.

Aggregate action shares are not fixed independently of sequencing because earlier actions change later feasibility. The frequency-informed diagnostic therefore does not reproduce the Q-policy’s realized action mix exactly. The relevant test is not whether Q uses stronger institutions, but whether its state-to-action mapping produces higher welfare than simpler multi-institution policies.

The Q-Policy Does Not Show a Statistically Distinguishable Advantage over Random Feasible Choice

The main independent adaptive comparison is learned Q versus uniform random feasible institutional choice (Fig. 4). Mean final planner welfare is 20.182 for learned Q and 20.046 for random feasible. Two additional ex-post diagnostics use information from Q behavior on the same evaluation games: frequency-informed random feasible reaches 20.315 and the shuffled learned sequence 20.044. The typical between-policy differences are therefore small relative to within-policy variation.

Relative to uniform random feasible choice, the Q-policy has a mean paired welfare difference of

$$\Delta\Phi_{Q-\text{random}} = +0.136 \quad (12)$$

with a seed-clustered 95% confidence interval of

$$[-0.336, 0.608]$$

The paired difference is also small and imprecisely estimated within each behavioral scenario. Across the six 100-seed scenario subsets, Q minus random feasible

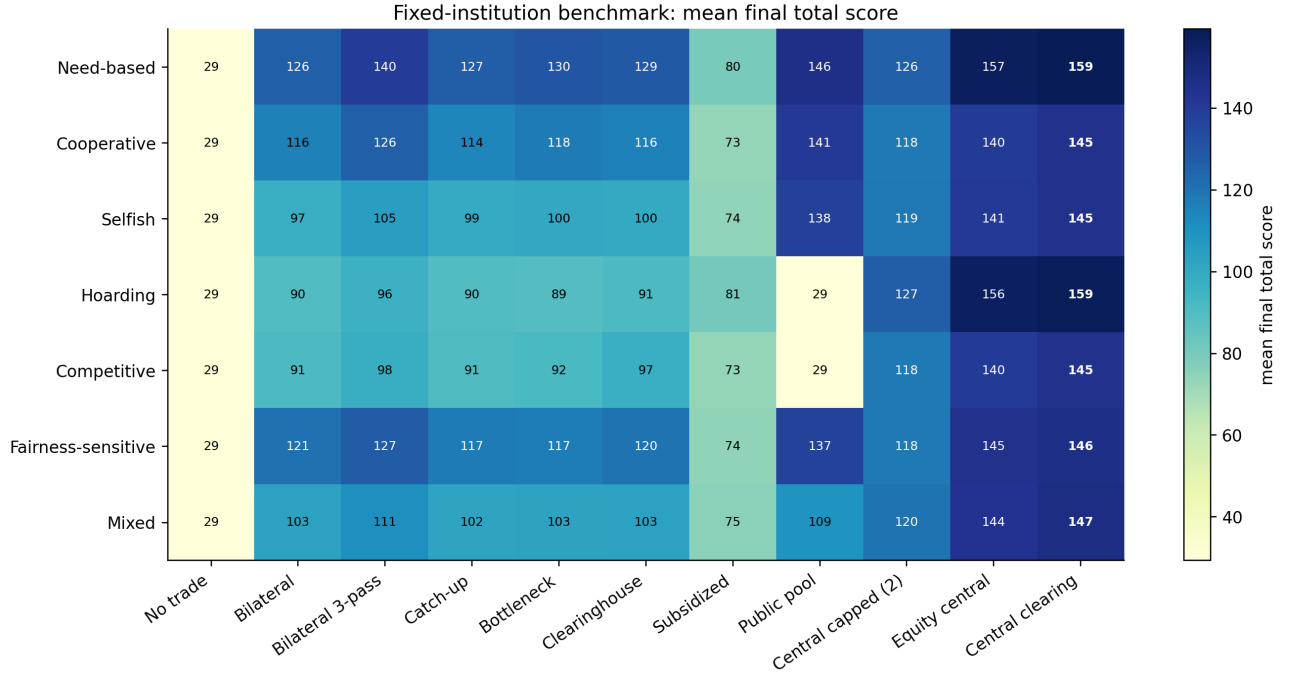


Figure 1: Final total score under fixed exchange institutions. The fixed experiment crosses two build modes, seven behavioral populations and eleven institutions over 200 matched seeds. The heatmap focuses on the development-oriented slice for readability, while the text reports the pooled build-mode contrast separately.

ranges from -0.328 welfare points in the cooperative-to-hoarding scenario to $+0.597$ in the need-based scenario. All six 95% paired percentile-bootstrap intervals include zero. Neither hidden regime-shift scenario shows a statistically distinguishable Q advantage (Table S.10).

Relative to the frequency-informed random feasible policy, the difference is

$$\Delta\Phi_{Q\text{-frequency}} = -0.133 \quad (13)$$

with a 95% confidence interval of

$$[-0.597, 0.324]$$

The shuffled learned sequence produces a difference of

$$\Delta\Phi_{Q\text{-shuffled}} = +0.138 \quad (14)$$

with a 95% confidence interval of

$$[-0.295, 0.555]$$

All three pooled intervals include zero. Because the frequency weights and shuffled sequence are constructed from Q behavior on these same evaluation games, they are not independent predictive baselines. Taken together, the independent random-feasible comparison and the two ex-post diagnostics do not establish that the Q-table’s state-dependent mapping adds welfare beyond simple multi-institution choice.

The policy therefore produces a structured action sequence without demonstrating a welfare gain attributable to the learned state-to-action mapping. The confidence interval includes zero and remains compatible with moderate advantages in either direction, so the result is inconclusive rather than evidence of equivalence. In practical terms, this trained Q-table provides little affirmative evidence that its state mapping improves welfare over simple feasible multi-institution choice.

Counterfactual Rollouts Show Continuation-Dependent First-Action Value Differences The branch diagnostic asks whether alternative feasible first actions produce different downstream welfare when they are evaluated from exactly the same checkpoint. Its independent selection/evaluation split avoids defining an apparent best action and estimating its advantage on the same noisy rollouts.

Under continuation with the Q-policy, the action selected as best on the 20 selection rollouts exceeds the independently evaluated second-selected action by 0.906 welfare points on average, with a base-game-clustered 95% confidence interval of $[0.559, 1.299]$. The selected action also exceeds the actual Q-chosen action by 1.213 points, with a 95% confidence interval of $[0.594, 1.978]$. The corresponding unadjusted checkpoint-level intervals exclude zero at 32 of 54 and 22 of 54 checkpoints, respectively.

Permanent-bilateral continuation produces similarly

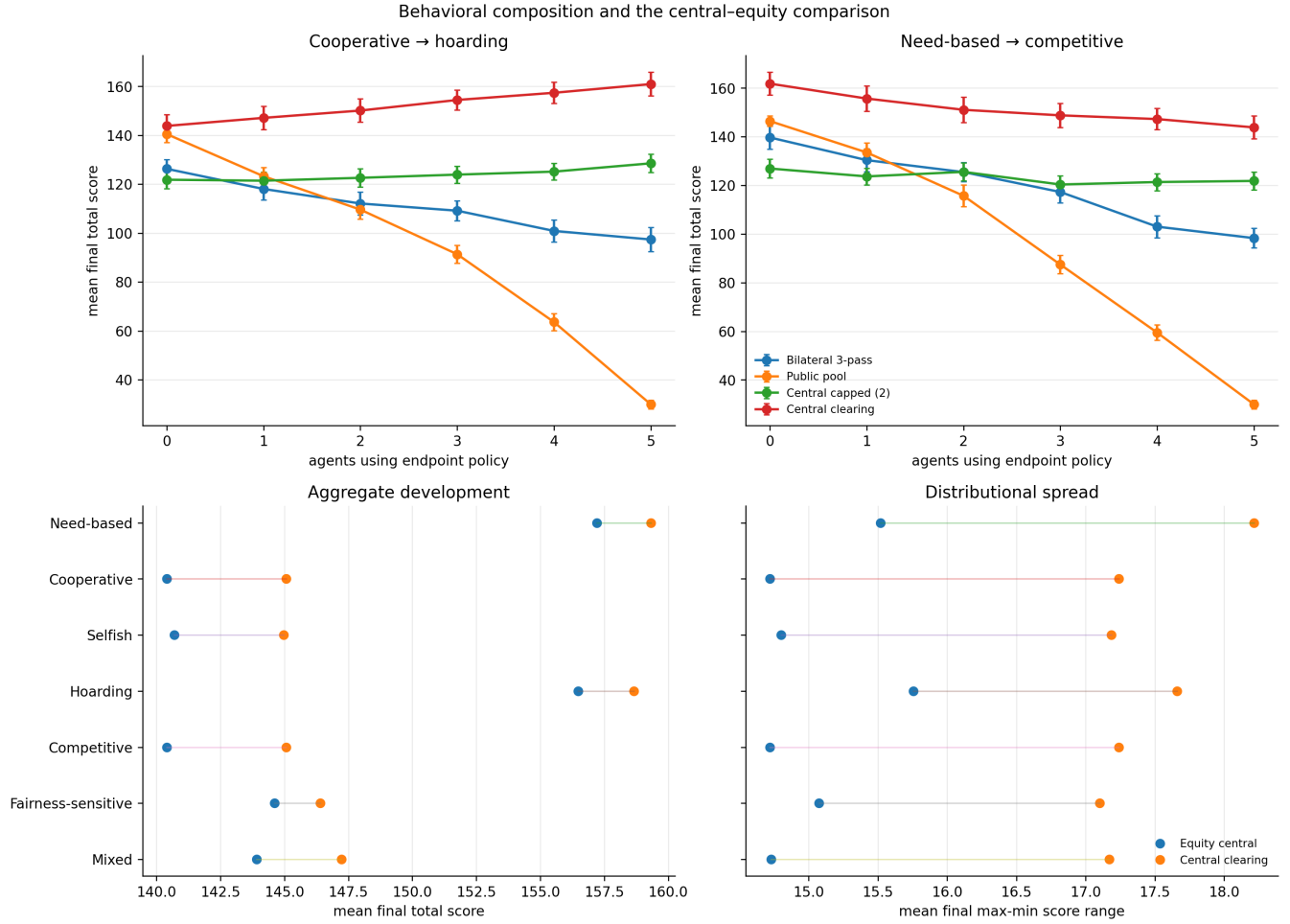


Figure 2: Behavioral composition and the output-distribution comparison. The composition sweep illustrates how institutional performance changes along the cooperative-to-hoarding and need-based-to-competitive gradients. The central-versus-equity contrast uses development-oriented building and reports both final total score and final max-min score range, defined as maximum minus minimum individual score.

positive aggregate gaps. Selected-best exceeds selected-second by 0.989 points on average, with a clustered interval of $[0.711, 1.269]$ and exceeds the Q-chosen action by 1.283 points, with an interval of $[0.695, 1.918]$. The unadjusted checkpoint-level intervals exclude zero at 27 of 54 and 20 of 54 checkpoints, respectively.

Random feasible continuation also produces positive aggregate differences. Selected-best exceeds selected-second by 0.594 points on average, with a clustered interval of $[0.348, 0.862]$ and exceeds Q-chosen by 0.696 points, with an interval of $[0.396, 1.093]$. None of the selected-best-minus-second checkpoint-level intervals and 1 of 54 selected-best-minus-Q-chosen intervals exclude zero. With only 20 evaluation continuations per action at each checkpoint, this low checkpoint-level resolution may reflect limited precision and is not evidence that the underlying rankings reverse or are intrinsically unstable.

The independent evaluation half therefore contains

aggregate first-action signal under all three continuation rules, but its magnitude and state-level precision depend on what policy follows the first action. Because the aggregate estimates rest on only 18 independent base games and the checkpoint intervals are descriptive and unadjusted, the diagnostic does not establish a sufficiently precise general state-specific ranking. It also should not be interpreted as a direct estimate of an intertemporal timing effect: it compares the value of different next actions conditional on a specified continuation policy.

Discussion

Institutional performance in the simulated economy is strongly conditional on the interaction between behavioral policy packages and exchange protocols. Once stronger coordination is made scarce, the tested tabular Q-policy does not demonstrate that its state-dependent

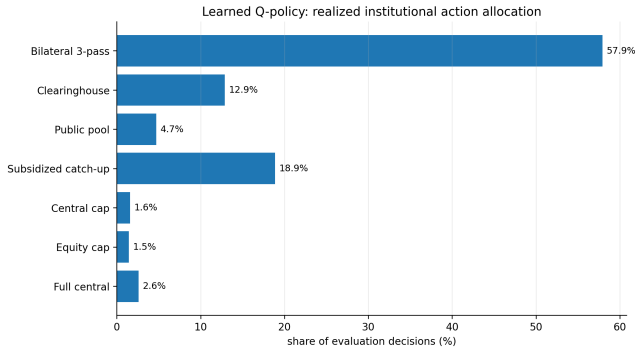


Figure 3: Realized action allocation of the Q-policy. The Q-policy uses bilateral exchange most often and stronger coordination mechanisms selectively. Because the coordination-cost system is deliberately constructed to make expensive actions scarce, this distribution is descriptive and is not by itself evidence that the learner discovered a good institutional strategy.

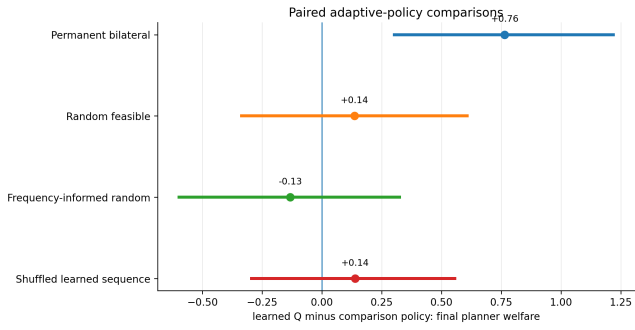


Figure 4: Paired welfare differences between the Q-policy and comparison policies. Random feasible is the independent multi-institution comparator. Frequency-informed random feasible and the shuffled sequence are ex-post diagnostics constructed from Q behavior on the same evaluation games. Error bars show seed-clustered 95% confidence intervals.

sequencing improves welfare over simple feasible multi-institution choice, even though counterfactual rollouts indicate that the value of the next institutional action can differ. These findings are limited to the modeled environment, but they suggest broader questions about how institutional rules expose behavioral heterogeneity and how adaptive institutional policies should be evaluated.

One interpretation of the fixed experiments is that exchange institutions can be characterized not only by how centralized they are, but also by which decisions they leave to individual agents. This interpretation is consistent with experimental evidence that institutional outcomes depend on behavioral responses to the rules participants face [Kosfeld et al., 2009, Fehr et al., 2021]. Bilateral bargaining preserves agent-generated offers and acceptance decisions, public pooling depends on voluntary contribution behavior while central clearing removes some of these individual veto points by

constructing reciprocal exchanges institutionally. Behavioral heterogeneity therefore has more opportunities to affect allocation under some institutional interfaces than under others. Because the behavioral profiles differ in project valuations and other decision parameters, the experiment cannot isolate the causal contribution of any single interface. Nevertheless, the results are consistent with the idea that institutional performance can depend on the behavioral decision points through which heterogeneous preferences influence allocation.

The central-versus-equity comparison raises a second issue. There is no institution-independent definition of the “best” outcome in the model. Standard central coordination performs better on aggregate development, whereas equity-oriented coordination performs better on the selected distributional criterion. The institutional ranking therefore depends heavily on the objective used to evaluate the resulting allocation. This becomes especially important in the adaptive setting: a learning algorithm does not discover an institution that is optimal independently of normative assumptions. In learning-based institutional design more generally, the choice of objective is therefore part of the design problem rather than a neutral technical input [Koster et al., 2025]. Here I opted for a combined measure of development and equity while still favoring development overall, though the distribution of weights could reasonably be different.

The adaptive experiment additionally provides a methodological lesson. A learned policy can generate structured and apparently plausible sequences of institutional choices without demonstrating that its state-dependent sequencing adds value. Comparing the Q-policy only with permanent bilateral exchange would make the learned policy appear relatively successful, but that comparison confounds access to a richer action set with useful state-dependent adaptation. Against uniform random feasible choice, which can also select among multiple institutions, no statistically distinguishable Q advantage is observed. The frequency-informed and shuffled diagnostics lead to the same cautious interpretation. This suggests that adaptive policies should be evaluated against comparators that separate the benefit of having multiple available interventions from the additional value of deciding when to deploy them.

The rollout diagnostic adds an important qualification. Alternative first actions do produce different continuation-conditioned values on average at the sampled checkpoints, including under random feasible continuation. The absence of a clear Q advantage therefore does not imply that institutional choice is irrelevant. Instead, the results are consistent with a harder problem: useful state-dependent variation may exist while remaining difficult for the compressed policy representation to exploit reliably. Partial observability and state aggregation are plausible explanations [Jaakkola et al., 1994, Singh et al., 1994], but the current experiment

cannot identify a single cause. The rollout results instead show action-value variation at the sampled checkpoints.

Limitations The environment remains deliberately stylized and small in terms of agent count, number of resources, game length and other design aspects. The fixed experiments do not isolate individual institutional dimensions or measured psychological traits.

The adaptive benchmark is likewise conditional on one synthetic coordination-capacity system and welfare function. Coordination stock, regeneration, institutional costs, switching rules and the affordability requirement were chosen to create an intertemporal scarcity problem rather than calibrated empirically. The planner is a coarse, partially observed tabular baseline trained once, so training-run variability is unknown and low unseen-state frequency does not establish adequate state-action visitation. The evaluation set is disjoint from training but was subsequently used when comparing additional learner variants. Random feasible is an independent but intentionally simple comparator while the frequency-informed and shuffled policies are ex-post diagnostics only. Finally, the branch analysis remains exploratory because its checkpoints are nested within only 18 independent base games and its estimated first-action values depend on the continuation policy.

Future Work Before extending the framework substantively, the present benchmark could be strengthened through sensitivity analysis of the coordination parameters and welfare weight, repeated planner training on a fresh test set, stronger state-dependent baselines and behavioral or institutional ablations that separate the package-level mechanisms observed here.

A broader direction is to develop the current environment into a much richer computational exchange economy. Increasing the number of agents, resources and build targets would allow coordination burdens to arise more directly from the scale of information collection, matching and transfer execution rather than primarily from fixed capacity costs. This could turn the current synthetic coordination constraint into a more endogenous trade-off between the allocation advantages of broader coordination and the burden required to implement it.

A further extension could introduce a monetary layer. The present economy relies on reciprocal resource exchange, so successful trade depends strongly on matching agents who simultaneously hold resources valued by one another. Money would relax that coincidence requirement by allowing agents to sell resources to one participant and use the resulting purchasing power elsewhere. A larger model with monetary balances and endogenous resource prices could therefore examine how monetary exchange changes the coordination problems created by resource complementarity and whether tar-

geted redistribution or centralized matching continues to add value once a common medium of exchange is available.

Conclusion This paper develops a controlled computational environment for studying how exchange institutions interact with heterogeneous decision policies under resource interdependence. The fixed experiments show that institutional outcomes depend strongly on the behavioral interfaces preserved by different protocol packages and that institutional rankings can change with the objective used to evaluate them. The adaptive extension then shows that introducing a scarce coordination resource creates a sequential choice problem, but also that a structured learned policy should not be credited with useful institutional timing merely because it produces complex action sequences.

The contribution is not a quantitative claim about real economies or a general theory of optimal institutional design. It should instead be understood as a proof of concept for a modular computational approach in which behavioral rules, exchange institutions and adaptive coordination can be varied and evaluated within the same environment. At this scale, the model already exposes two issues that become more important rather than less in richer systems: institutions differ in where they permit heterogeneous behavior to affect allocation and adaptive institutional policies require demanding evaluation before apparent structure can be interpreted as successful learning. Extending the framework to larger exchange systems, endogenous coordination burdens and monetary transactions would provide a natural route from this small-scale pilot toward broader questions about how resource, monetary and institutional systems interact.

References

- Douglass C. North. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, Cambridge, 1990.
- R. H. Coase. The problem of social cost. *The Journal of Law and Economics*, 3:1–44, 1960. doi: 10.1086/466560.
- Ernst Fehr and Klaus M. Schmidt. A theory of fairness, competition, and cooperation. *The Quarterly Journal of Economics*, 114(3):817–868, 1999. doi: 10.1162/003355399556151.
- Gary E. Bolton and Axel Ockenfels. ERC: A theory of equity, reciprocity, and competition. *American Economic Review*, 90(1):166–193, 2000. doi: 10.1257/aer.90.1.166.
- Gary Charness and Matthew Rabin. Understanding social preferences with simple tests. *The Quarterly Journal of Economics*, 117(3):817–869, 2002. doi: 10.1162/003355302760193904.
- Urs Fischbacher, Simon Gächter, and Ernst Fehr. Are people conditionally cooperative? evidence from a public goods experiment. *Economics Letters*, 71(3):397–404, 2001. doi: 10.1016/S0165-1765(01)00394-9.

- Leonid Hurwicz. The design of mechanisms for resource allocation. *American Economic Review*, 63(2):1–30, 1973.
- Eric S. Maskin. Mechanism design: How to implement social goals. *American Economic Review*, 98(3):567–576, 2008. doi: 10.1257/aer.98.3.567.
- Leigh Tesfatsion. Agent-based computational economics: A constructive approach to economic theory. In Leigh Tesfatsion and Kenneth L. Judd, editors, *Handbook of Computational Economics*, volume 2, pages 831–880. Elsevier, 2006. doi: 10.1016/S1574-0021(05)02016-2.
- Joshua M. Epstein. Agent-based computational models and generative social science. *Complexity*, 4(5):41–60, 1999. doi: 10.1002/(SICI)1099-0526(199905/06)4:5(41::AID-CPLX9)3.0.CO;2-F.
- Kevin R. McKee, Andrea Tacchetti, Michiel A. Bakker, Jan Balaguer, Lucy Campbell-Gillingham, Richard Everett, and Matthew Botvinick. Scaffolding cooperation in human groups with deep reinforcement learning. *Nature Human Behaviour*, 7(10):1787–1796, 2023. doi: 10.1038/s41562-023-01686-7.
- Raphael Koster, Miruna Pislari, Andrea Tacchetti, Jan Balaguer, Leqi Liu, Romuald Elie, Oliver P. Hauser, Karl Tuyls, Matt Botvinick, and Christopher Summerfield. Deep reinforcement learning can promote sustainable human behaviour in a common-pool resource problem. *Nature Communications*, 16(1):2824, 2025. doi: 10.1038/s41467-025-58043-7.
- Andrea Tacchetti, Raphael Koster, Jan Balaguer, Leqi Liu, Miruna Pislari, Matthew M. Botvinick, Karl Tuyls, David C. Parkes, and Christopher Summerfield. Deep mechanism design: Learning social and economic policies for human benefit. *Proceedings of the National Academy of Sciences*, 122(25):e2319949121, 2025. doi: 10.1073/pnas.2319949121.
- Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, Cambridge, 1990. doi: 10.1017/CBO9780511807763.
- Elinor Ostrom, James Walker, and Roy Gardner. Covenants with and without a sword: Self-governance is possible. *American Political Science Review*, 86(2):404–417, 1992. doi: 10.2307/1964229.
- Michael Kosfeld, Akira Okada, and Arno Riedl. Institution formation in public goods games. *American Economic Review*, 99(4):1335–1355, 2009. doi: 10.1257/aer.99.4.1335.
- Ernst Fehr, Michael Powell, and Tom Wilkening. Behavioral constraints on the design of subgame-perfect implementation mechanisms. *American Economic Review*, 111(4):1055–1091, 2021. doi: 10.1257/aer.20170297.
- Christopher J. C. H. Watkins and Peter Dayan. Q-learning. *Machine Learning*, 8:279–292, 1992. doi: 10.1007/BF00992698.
- Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, Cambridge, MA, 2 edition, 2018.
- Tommi Jaakkola, Satinder P. Singh, and Michael I. Jordan. Reinforcement learning algorithm for partially observable markov decision problems. In *Advances in Neural Information Processing Systems 7*, pages 345–352. MIT Press, 1994.
- Satinder P. Singh, Tommi Jaakkola, and Michael I. Jordan. Reinforcement learning with soft state aggregation. In *Advances in Neural Information Processing Systems 7*, pages 361–368. MIT Press, 1994.

Supplementary Materials

A Details on Methods

A.1 Coordination-Capacity Costs

The adaptive planner must be able to afford the maximum possible cost of an institution before selecting it. Realized cost is

$$c_t(a) = b(a) + s_t(a) + w_t(a)$$

where $b(a)$ is the base burden, $s_t(a)$ is the switching cost and $w_t(a)$ is workload cost. Entering a non-bilateral regime from a different previous action costs one unit. Starting the game, remaining in the same regime or returning to bilateral exchange costs zero.

For non-bilateral actions, realized workload is

$$w_t(a) = \min \left(w_{\max}(a), \left\lceil \frac{u_t(a)}{d(a)} \right\rceil \right)$$

where $u_t(a)$ is the observed workload in resource units and $d(a)$ is the number of workload units required for one capacity-cost unit. Bilateral three-pass has $w_t = 0$. Executed reciprocal exchanges count the offered plus requested resource units. Subsidies count transferred quantity and public-pool workload counts contributions, allocations and returns.

Table S.1: Coordination-capacity parameters. The ex-ante commitment check uses base cost, switching cost and the maximum workload cost. After execution, only realized workload is charged. “Units/cost” is the number of workload units required to add one realized capacity-cost unit.

Institution	Base	Units/cost	Max workload
Bilateral 3-pass	0	–	0
Clearinghouse	1	6	1
Public pool	1	6	2
Subsidized catch-up	1	2	1
Central cap	2	4	1
Equity cap	2	4	1
Full central clearing	3	6	2

The coordination stock is initialized at $G_{\max} = 8$. After the institution and build phases, realized institutional cost is deducted and one unit of capacity is recovered, capped at $G_{\max}$. The cost values are modeling parameters, not empirical estimates of governance or administrative cost.

A.2 Q-Learning Configuration and Evaluation Details

The state representation has seven discrete features:

- **Phase:** $\min(3, \lfloor 4t/T \rfloor)$, giving four game-phase bins.
- **Coordination capacity:** zero, positive capacity at or below one third of maximum, above one third and at or below two thirds or above two thirds.
- **Bottleneck severity:** high if total missing units are at least 11 or at least five agents are blocked, medium if missing units are at least 8 or at least three agents are blocked and otherwise low.
- **Voluntary-exchange viability:** no requests, match rate below 0.34, match rate below 0.67 or at least 0.67.
- **Score-spread category:** max–min score difference below 7, below 14 or at least 14.
- **Idle infrastructure:** below 5, below 10 or at least 10 unused infrastructure units.
- **Previous action:** the code of the institution used in the previous round.

Table S.2: Configuration of the tabular Q-learning planner. The table reports the main parameters of the learner used for the final policy evaluation.

Parameter	Value
Training episodes	12,000
Training seeds	100000–111999
Rounds per game	20
Q initialization	zeros for newly visited states
Initial learning rate	0.25
Learning-rate floor	0.01
Learning-rate rule	$0.25/\sqrt{n}$, floored at 0.01
Discount factor γ	1.0
Initial exploration ϵ	1.0
Exploration floor	0.03
Exploration schedule	quadratic decay over first 90% of training
Training tie-break	random among equal-Q feasible actions
Evaluation tie-break	stable action order
Planner welfare weight λ	0.25
Evaluation seeds	500000–500099
Evaluation exploration	0
Evaluation learning	disabled

The fixed action order is bilateral three-pass, clearinghouse, public pool, subsidized catch-up, central cap, equity cap and full central clearing. Exploitation and one-step or n-step bootstrapping maximize over currently feasible actions only. Terminal transitions use immediate reward without a future bootstrap term.

The six adaptive training/evaluation scenarios are need-based, cooperative, hoarding, mixed, need-based-to-competitive and cooperative-to-hoarding. The first three are homogeneous five-agent populations. The mixed scenario contains one cooperative, selfish, hoarding, competitive and fairness-sensitive agent, with assignment shuffled by seed. The two transition scenarios are synthetic regime-shift stress tests rather than empirical behavioral trajectories: all five agents begin as need-based or cooperative, respectively and at the start of simulation round index 10—the second half of the 20-round game—all five switch to competitive or hoarding policies. They expose the baseline planner to abrupt within-game changes while keeping the scenario set compact. The six scenarios are not intended to cover the complete seven-population fixed design. During the 12,000 training episodes, one of the six scenarios is sampled uniformly at random per episode using a fixed training RNG. The realized training counts are 1,944 cooperative, 1,977 cooperative-to-hoarding, 2,038 need-based, 2,028 need-based-to-competitive, 2,009 hoarding and 2,004 mixed episodes.

The evaluation set is disjoint from Q training but was later reused to compare additional Q-learning variants. The Q-policy makes 12,000 evaluation decisions over 100 seeds and six scenarios. Twenty-one decisions (0.175%) encounter a state not visited during training. All select bilateral three-pass because unseen states are read as zero-valued Q entries and the stable tie-break selects the first feasible action in the fixed action order.

Pooled policy uncertainty is clustered by the 100 evaluation seeds because each seed is reused across six scenarios. The production RNG is separate from institution and policy RNG streams, so policy-specific random choices do not consume subsequent production draws.

A.3 Realized Action Shares in Policy Evaluation

Table S.3: Realized pooled action shares of the learned Q-policy. Shares are rounded to one decimal place to match the regenerated action-allocation figure.

Policy	Bilat.	Clear.	Pool	Subsidy	Cap	Equity	Full
Learned Q	57.9%	12.9%	4.7%	18.9%	1.6%	1.5%	2.6%

A.4 Behavioral Profiles

The fixed experiment uses the following behavioral populations and policy classes:

These names are model labels rather than measured psychological categories. All six homogeneous classes use the same utility-based architecture, but their weights can alter project targeting as well as exchange valuation. Need-based agents request missing resources for their current target and trade only from spare stock. Cooperative policies use lower transaction thresholds. Selfish and hoarding policies protect progressively larger reserves while competitive and fairness-sensitive policies additionally modify acceptance according to relative scores. Tables S.5 and S.6 report

Table S.4: Behavioral policy populations in the fixed experiment. All homogeneous policies use the same decision architecture and feasible project set within the selected build mode, but differ in utility weights affecting project targeting and exchange valuation as well as direct exchange and pooling parameters.

Population	Policy class	Main policy tendency
Need-based	NeedBasedTradePolicy	Target bottlenecks
Cooperative	CooperativeTradePolicy	Readier exchange/contribution
Selfish	SelfishTradePolicy	Stronger own-target reserves
Hoarding	HoardingTradePolicy	High surplus/terms requirements
Competitive	CompetitiveTradePolicy	Avoid helping stronger rivals
Fairness-sensitive	FairnessSensitiveTradePolicy	Balance/inequality heuristics
Mixed	five policy classes	One of each non-need-based type

Table S.5: Direct exchange parameters of the homogeneous behavioral policies. All policies request at most two units at a time. “Reserve” is the additional stock held above the current target reserve. Δ_{offer} and Δ_{accept} are minimum utility changes for proposing and accepting a trade. Competitive acceptance requires an additional 0.10 utility when the responder is ahead in score. Fairness-sensitive acceptance lowers the threshold by 0.05 when trading with a lower-scoring responder and raises it by 0.10 when trading with a higher-scoring responder.

Policy	Reserve	Max payment	Δ_{offer}	Δ_{accept}
Need-based	0	4	0.00	-0.05
Cooperative	0	4	-0.10	-0.10
Selfish	1	4	0.05	0.00
Hoarding	2	5	0.25	0.10
Competitive	1	4	0.10	0.05
Fairness-sensitive	0	4	-0.05	-0.05

Table S.6: Utility weights used by the homogeneous behavioral policies. The common utility system uses weights on own score (S), social welfare (W), relative advantage (R), equity (E), productive capacity (P), site progress (T), innovation preference (I), security (H), target progress (G) and crown incentives (C). These weights enter project-target/build scoring and exchange-related utility calculations. Security also affects stock protection and pool contribution behavior while the crown term is active only in crown-aware build mode. These values are design parameters rather than empirical estimates.

Policy	S	W	R	E	P	T	I	H	G	C
Need-based	1.50	0.20	0.20	0.00	2.00	2.40	0.10	0.25	1.00	0.30
Cooperative	1.00	1.00	0.00	0.20	1.80	2.00	0.30	0.15	0.90	0.20
Selfish	1.60	0.00	0.30	0.00	1.80	2.00	0.15	0.35	1.10	0.30
Hoarding	1.10	0.00	0.10	0.00	1.50	1.70	0.00	1.00	1.20	0.15
Competitive	1.50	-0.10	1.00	0.00	1.70	1.90	0.20	0.40	1.00	0.40
Fairness-sensitive	1.00	0.40	0.00	1.00	1.70	1.80	0.15	0.25	0.95	0.15

the numerical parameters that define the six homogeneous rule classes.

A.5 Resource Access and Leader Bonuses

For every fixed-experiment seed, five balanced asymmetric access profiles are generated and reused across all compared conditions. Each resource is primary for exactly one agent and secondary for exactly one agent. Primary access has weight 0.45, secondary access 0.25 and each of the remaining three resources 0.10. Production samples from normalized effective weights. Agents begin with two resource draws per round. Production sites add one draw and advanced sites contribute two with total draws capped at seven.

A production-site build adds 0.10 to the agent’s currently weakest effective access weight. Ties use the fixed resource order. The bonus changes subsequent normalized production probabilities but is assigned automatically rather than chosen strategically.

The model also has two transferable three-point leader bonuses. The infrastructure bonus requires a unique leader with at least five infrastructure units and the innovation bonus a unique leader with at least three innovations. An incumbent retains a bonus during a tie and loses it only when another agent moves strictly ahead. Leader status is updated immediately after each successful build. Under development-oriented building, project targets are scored by the policy-specific utility weights without a crown bonus. Crown-aware building adds the crown term, which can redirect the same utility-based target selection toward near-term stealing, defense or claiming of these leader bonuses.

A.6 Build Types and Resource Requirements

Production takes place before exchange and building in each round, so productive capacity constructed during a round affects later rounds rather than producing retroactively in the current round.

Table S.7: Build projects in the simulation. Production sites also add 0.10 to the agent’s currently weakest effective resource-access weight. An advanced site upgrades an existing production site rather than using an additional site slot.

Project	Resource cost	Points	Structural effect
Infrastructure	1 materials, 1 components	0	Every two infrastructure units support one site slot.
Production site	1 materials, 1 components, 1 food, 1 energy	3	Adds one production draw and an access bonus.
Advanced site	2 energy, 3 knowledge	5	Upgrades one production site and raises its production contribution from one to two draws.
Innovation	1 food, 1 energy, 1 knowledge	1 for each of first three	Counts toward innovation leadership even after direct scoring stops.

The fixed experiment also includes transferable infrastructure and innovation leader bonuses. In development-oriented mode, the utility-based build score contains no crown-specific term. Crown-aware mode activates that additional term, allowing near-term crown stealing, defense or claiming to alter the project selected by an otherwise identical behavioral policy.

A.7 Institution Definitions

Table S.8 summarizes all eleven institutions in the fixed experiment. The mechanisms deliberately differ in more than one dimension: some preserve voluntary offer and acceptance decisions, while others replace those decisions with institutional matching or redistribution.

Table S.8: Institutions in the fixed experiment. “Hooks” indicates the main behavioral exchange decisions that remain active. Central clearing is greedy reciprocal matching rather than globally optimal or price-clearing allocation.

Institution	Operational rule	Behavioral hooks
No trade	No cross-agent transfer.	None
Bilateral	One request–offer–accept bargaining pass. The cheapest acceptable feasible offer executes for each processed request.	Requests, offers, acceptance
Bilateral 3-pass	Repeats the same bilateral pipeline for up to three passes, stopping early after a pass with no trade.	Requests, offers, acceptance
Catch-up bilateral	Bilateral bargaining with lower-scoring requesters processed first.	Requests, offers, acceptance
Bottleneck bilateral	Bilateral bargaining with requests prioritized by how strongly receipt would reduce the requester’s target shortage.	Requests, offers, acceptance
Clearinghouse	Collects policy-generated requests and accepted offers, then greedily executes a compatible subset from the global offer set.	Requests, offers, acceptance
Subsidized catch-up	Processes low-scoring agents. An agent exactly one unit short of its target may receive that unit directly from a donor with at most two subsidies per round.	Target/reserve information with no barter acceptance
Public pool	Collects voluntary surplus contributions and allocates the round-local pool toward agents close to targets, prioritizing lower scores with at most three allocations per round.	Contribution and allocation acceptance
Central clearing	Uses agent requests, centrally constructs one-for-one reciprocal barter candidates, ranks them by target-shortage reduction and greedily executes feasible matches.	Requests only with offers/acceptance bypassed
Equity central	Uses the central-clearing procedure but gives additional priority to lower-scoring requesters.	Requests only with offers/acceptance bypassed
Central capped (2)	Uses central clearing but permits at most two successful matches per round.	Requests only with offers/acceptance bypassed

B Additional Results

B.1 Clustered Policy Comparisons

Table S.9: Pooled policy comparisons using seed-clustered 95% confidence intervals. Each comparison contains 600 seed–scenario observations nested within 100 evaluation-seed clusters. Positive values favor the first policy.

Comparison	Mean	CI low	CI high
Q – random feasible	0.136	-0.336	0.608
Q – frequency-informed	-0.133	-0.597	0.324
Q – shuffled sequence	0.138	-0.295	0.555

Random feasible is the independent multi-institution comparator. Frequency-informed and shuffled comparisons are ex-post diagnostics constructed from learned-Q behavior on the same evaluation games.

Table S.10: Scenario-specific paired Q minus random-feasible welfare differences. Each row contains 100 matched evaluation seeds. Intervals use 2,000-draw percentile bootstraps of the paired within-scenario differences. None excludes zero.

Scenario	Mean diff.	95% CI low	95% CI high
Need-based	0.597	-0.343	1.516
Cooperative	-0.165	-1.151	0.799
Hoarding	0.113	-0.926	1.097
Mixed	0.218	-0.590	0.977
Need-based → competitive	0.383	-0.441	1.232
Cooperative → hoarding	-0.328	-1.219	0.601

B.2 Counterfactual Branch Diagnostic

The diagnostic uses 18 independent base games, 54 checkpoints and 40 continuations per feasible candidate action. The first 20 continuations are used to select provisional best and second-best actions and the remaining 20 evaluate those preselected actions independently.

The aggregate estimates are based on only 18 independent base-game clusters. Positive aggregate gaps under random feasible continuation indicate some independently evaluated first-action signal at these checkpoints, while the low checkpoint-level resolution shows that the available 20 evaluation rollouts per action often do not identify a difference

Table S.11: Selected branch comparisons from the rollout diagnostic. Aggregate intervals are clustered at the 18-base-game level. “CI excludes 0” counts checkpoint-level evaluation-half intervals that exclude zero. These intervals are unadjusted for multiple comparisons and are descriptive.

Continuation	Comparison	Mean	95% CI	CI excludes 0
Learned Q	Best – second	0.906	[0.559, 1.299]	32/54
Learned Q	Best – Q-chosen	1.213	[0.594, 1.978]	22/54
Permanent bilateral	Best – second	0.989	[0.711, 1.269]	27/54
Permanent bilateral	Best – Q-chosen	1.283	[0.695, 1.918]	20/54
Random feasible	Best – second	0.594	[0.348, 0.862]	0/54
Random feasible	Best – Q-chosen	0.696	[0.396, 1.093]	1/54

precisely at an individual state. Because low power is not evidence of ranking instability, the mixed/inconclusive label reflects limited state-level precision, the small number of independent base games and dependence of the estimated contrasts on the continuation rule—not a claim that rankings are known to reverse.

B.3 Additional Q-Learning Variants

Several additional learner variants were tested alongside the main Q-learning specification.

Table S.12: Summary of attempted Q-learning extensions. None produced a stable advantage on the fixed evaluation set sufficient to replace the main learner. These experiments are therefore treated as robustness and diagnostic work rather than as additional main results.

Variant	Purpose	Outcome
State-v2	Add lower-tail information and modify tie-breaking	Some diagnostics improved, but policy-level comparisons remained unresolved.
Longer-return variants	Test n-step and Monte Carlo-style credit assignment	No stable policy-level improvement on the fixed evaluation set.
State-v3	Substantially enlarge the tabular state representation	Performance remained unstable and did not justify further expansion of the Q-table.